What does the quarkonium spectrum know about the heavy-quark potential?

A model-independent extraction of $V(r)$ from charmonium and bottomonium, and its consequences for B_c

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We determine the static heavy-quark potential $V(r)$ directly from the measured charmonium and bottomonium spectra, without assuming a functional form. $V(r)$ is represented by a cubic spline with up to twelve knots on $0.15 \le r \le 8\,\mathrm{GeV}^{-1}$ plus a Coulomb term, and is fitted together with the quark masses and a running coupling to 27 J-resolved $c\bar{c}$ and $b\bar{b}$ levels from the 2026 Review of Particle Physics, with a smoothness prior whose strength is scanned. The result is a potential with a one-standard-deviation band: $V(r)$ is determined to 30–50 MeV for $0.7 \lesssim r \lesssim 7\,\mathrm{GeV}^{-1}$, to 100–300 MeV for $0.2 \lesssim r \lesssim 0.5\,\mathrm{GeV}^{-1}$, and not at all below $0.2\,\mathrm{GeV}^{-1}$, where the spectrum has no sensitivity and the short-distance form is fixed only through the hyperfine splittings. The band is stable against the number of knots and the prior strength and the fit quality ($\chi^2 = 40$ for 27 levels) is better than that of any parametric form. Compared at equal shape, the logarithmic and power-law potentials lie within the band everywhere, whereas the Cornell potential is excluded at three standard deviations at $r \simeq 3\,\mathrm{GeV}^{-1}$. Propagated to $c\bar{b}$, the extraction gives B_c masses with uncertainties of 4–5 MeV that are independent of the choice of potential: $B_c(1S) = 6264 \pm 5$, $B_c^*(1S) = 6336 \pm 6$, $B_c(2S) = 6874 \pm 5$ MeV, in agreement with ATLAS for B_c^* and with the measured $2S$, but 10 MeV low for the ground state, so that the $2S$–$1S$ spacing of B_c is predicted 12 ± 7 MeV larger than measured. This is the only place where the $c\bar{b}$ system departs from the potential determined by $c\bar{c}$ and $b\bar{b}$, and it cannot be removed by any change of the potential's shape.

1 Introduction

Non-relativistic potential models describe the spectra of charmonium and bottomonium to a few tens of MeV, and have done so for fifty years with a handful of parametric forms [1–5]. The question we ask here is the inverse one: given the spectra, what is actually known about $V(r)$? Which distances are constrained, by which states, and how well? The question has a practical side. Predictions for states that have not been measured—the P and D waves of B_c, the higher bottomonia—are quoted with a "model dependence" estimated by comparing a few parametric potentials. That estimate is as arbitrary as the choice of potentials. A potential extracted from the data with a proper error band turns the model dependence into a derived quantity.

There is a well-known answer to a related question: the static potential computed in lattice QCD [6] and the phenomenological fits agree in the range $0.2 \lesssim r \lesssim 1\,\mathrm{fm}$, i.e. $1 \lesssim r \lesssim 5\,\mathrm{GeV}^{-1}$, where the potentials of Refs. [1–3] nearly coincide despite their different functional forms [7]. What has not been done, to our knowledge, is to let the spectrum itself determine $V(r)$ with current data, including the spin-dependent levels and the D waves, and to quantify the result with a covariance. That is the content of this paper.

Our method is a flexible (spline) representation of $V(r)$ fitted by least squares with a smoothness prior, which is the standard approach to ill-posed inverse problems and gives a covariance for the function. We apply it to the data set and spin-dependent treatment of a companion paper on B_c [8], so that the two analyses can be compared directly. Section 2 describes the parametrisation and the fit, Sec. 3 the extracted potential and its sensitivity, Sec. 4 the consequences for B_c, and Sec. 5 concludes.

2 Method

2.1 Flexible potential

We write

$$V(r) = -\frac{\kappa}{r} + S(r), \tag{1}$$

where S is a natural cubic spline in $x = \ln r$ through K knots log-spaced on $[0.15, 8]\,\mathrm{GeV}^{-1}$, the range occupied by the wave functions of the fitted states (from the inner 1% of the $b\bar{b}$ $1S$ to the outer 99% of the $c\bar{c}$ $3S$). Below the first knot S is held constant, so that the Coulomb term alone sets the short-distance behaviour; beyond the last knot S is continued linearly in r with its slope at $8\,\mathrm{GeV}^{-1}$, a string-like extrapolation that affects only states reaching beyond the data. An additive constant in V is exactly degenerate with $m_c + m_b$; we remove it by fixing the knot at $r = 1\,\mathrm{GeV}^{-1}$ so that $V(1\,\mathrm{GeV}^{-1}) = -1.166\,\mathrm{GeV}$, the value of the Martin potential of Ref. [9]. The free parameters are the $K - 1$ remaining knot values, κ, m_c, m_b and the QCD scale Λ of a one-loop running coupling $\alpha_s(\mu)$, $\mu = \sqrt{m_1 m_2}$, that enters the spin-dependent terms. With $K = 8$ there are eleven.

Bound states are obtained by integrating the radial Schrödinger equation with the Numerov method and locating eigenvalues by node-counting bisection; fine and hyperfine structure is added in first-order perturbation theory following Eichten and Quigg [9], with the $J = L$ states of unequal-mass mesons mixed. These are the methods of Ref. [8], where they are described in full; the B_c predictions below are therefore directly comparable to the parametric ones of that paper.

Table 1: The extracted potential (GeV) at selected r for the reference fit, with the three parametric potentials of Ref. [8] shifted to the same value at $r = 1\,\mathrm{GeV}^{-1}$.

r [GeV^{-1}]	$V \pm \sigma_V$	Martin	Log	Cornell
0.2	-2.504 ± 0.271	-2.192	-2.358	-3.136
0.3	-2.202 ± 0.129	-1.949	-2.057	-2.358
0.5	-1.761 ± 0.096	-1.629	-1.679	-1.714
0.75	-1.401 ± 0.041	-1.362	-1.379	-1.364
1	-1.166 ± 0.001	-1.166	-1.166	-1.166
1.5	-0.880 ± 0.047	-0.880	-0.866	-0.922
2	-0.688 ± 0.040	-0.670	-0.653	-0.754
3	-0.399 ± 0.029	-0.363	-0.353	-0.493
4	-0.173 ± 0.039	-0.138	-0.140	-0.271
6	$+0.153 \pm 0.036$	$+0.191$	$+0.161$	$+0.136$
8	$+0.383 \pm 0.080$	$+0.433$	$+0.374$	$+0.524$

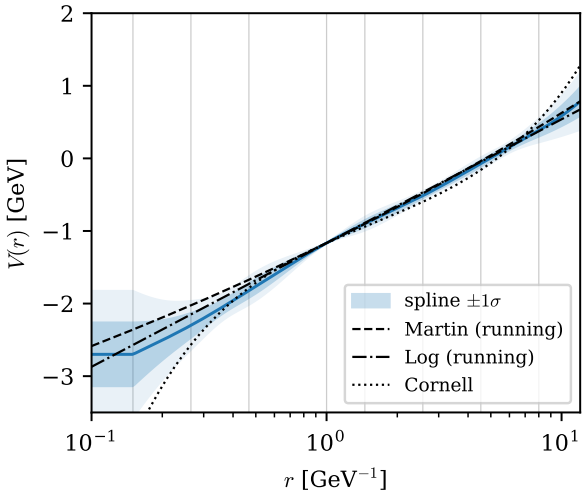


Figure 1: The potential extracted from the $c\bar{c}$+$b\bar{b}$ spectrum ($K = 8$, $\lambda = 1$): central value with 1σ (dark) and 2σ (light) bands. The refitted Martin, logarithmic and Cornell potentials of Ref. [8], shifted by a constant to $V(1\,\mathrm{GeV}^{-1}) = -1.166\,\mathrm{GeV}$, are shown for comparison; vertical lines mark the knots.

2.2 Data and fit

The data are the 27 J-resolved charmonium and bottomonium levels below open-flavour threshold listed in Ref. [8] (Table 1 there), taken from the 2026 Review of Particle Physics [10]: η_c, J/ψ, the four $1P$ states, $\eta_c(2S)$, $\psi(2S)$, $\psi(3770)$, $\psi_2(3823)$ and $\psi_3(3842)$ for charmonium; η_b, $\Upsilon(1S)$, the four $1P$ states, $\eta_b(2S)$, $\Upsilon(2S)$, $\Upsilon_2(1D)$, the four $2P$ states, $\Upsilon(3S)$, $\chi_{b1}(3P)$ and $\chi_{b2}(3P)$ for bottomonium. A theory floor of 10 MeV is added in quadrature to the experimental errors.

The residual vector is the 27 data residuals followed by the smoothness prior

$$\sqrt{\lambda}\,\frac{s_{k+1} - 2s_k + s_{k-1}}{(\Delta x)^2}, \qquad k = 2, \ldots, K-1, \quad (2)$$

on the knot values s_k, and the sum of squares is minimised with a trust-region least-squares algorithm. The covariance of the parameters is the inverse Gauss–Newton matrix of the full residual vector, so the prior is included in the error band, as it should be: the band quantifies what the data plus the smoothness assumption determine. The band on $V(r)$ follows from the covariance of the knot values by linear propagation (the spline is linear in them), and vanishes at $r = 1\,\mathrm{GeV}^{-1}$ by construction; everywhere else it measures the uncertainty of the shape relative to that point.

The prior strength λ and the number of knots K are not fitted. We scan $K \in \{6, 8, 10, 12\}$ and $\lambda \in \{0.1, 1, 10, 100\}$ and require that the extracted band be stable; we also repeat the reference fit with the $c\bar{c}$ D waves or the $b\bar{b}$ $3P$ levels removed, with a 5 MeV floor, and with κ fixed.

3 The extracted potential

Figure 1 and Table 1 show the result for the reference fit ($K = 8$, $\lambda = 1$), which has $\chi^2 = 41.5$ for the 27 levels with a prior contribution of 2.0. This is a better description of the spectrum than any parametric form we have tried: the best of the three families in Ref. [8], the logarithmic potential with running coupling, has $\chi^2 = 59$ for the same data, and the Martin potential 70. The improvement is not an artefact of the extra freedom: it is the same ($\chi^2 = 40$–42) for $K = 6$, 8, 10 and 12 (Table 2), so the data determine a definite shape that six knots already capture and twelve do not over-fit.

Where the potential is known. Figure 2 shows the band width. $V(r)$ is determined to better than 50 MeV over $0.7 \le r \le 6.9\,\mathrm{GeV}^{-1}$ and to 30–40 MeV over most of that range; the width grows to 100 MeV at $0.3\,\mathrm{GeV}^{-1}$ and 300 MeV at $0.2\,\mathrm{GeV}^{-1}$, and the potential is unconstrained below $0.15\,\mathrm{GeV}^{-1}$. The sensitivity matrix $\partial E_i/\partial s_k$ (Fig. 3) makes the origin of this explicit. Every level responds mainly to the two or three knots nearest its mean radius: the $b\bar{b}$ $1S$ to $0.5\,\mathrm{GeV}^{-1}$, the $b\bar{b}$ $1P$ and $c\bar{c}$ $1S$ to $1.5\,\mathrm{GeV}^{-1}$, the $c\bar{c}$ $1P$ and $b\bar{b}$ $2S$, $1D$, $2P$ to $2.6\,\mathrm{GeV}^{-1}$, the $c\bar{c}$ $2S$, D waves and $b\bar{b}$ $3S$, $3P$ to $4.5\,\mathrm{GeV}^{-1}$. No level has more than 3% sensitivity to the knots at 0.15 and $0.26\,\mathrm{GeV}^{-1}$. The quarkonium spectrum is blind to the Coulomb region.

The short-distance region and the hyperfine splittings. That blindness has a consequence for κ. In the reference fit κ runs to zero, and fixing it to 0.3 or 0.5 (the one-gluon-exchange value $\frac{4}{3}\alpha_s$ for $\alpha_s \simeq 0.25$–0.4) worsens χ^2 to 47 and 51 while leaving $V(r)$ for $r > 0.5\,\mathrm{GeV}^{-1}$ unchanged within the band (Table 2). The spectrum is not pushing κ down—it cannot see it; the hyperfine splittings are. In first-order perturbation theory the S-wave hyperfine splitting is proportional to $|R(0)|^2$, which a Coulomb singularity enhances, and the fit lowers κ to reduce the over-predicted $c\bar{c}$ splitting. This is the same tension between the $c\bar{c}$ and

Table 2: Stability of the extraction. For each fit: data χ^2 (27 levels), prior χ^2, band width σ_V (MeV) at four distances, and central V (GeV) at three. Variants: no D ($c\bar{c}$ D waves removed), no $3P$ ($b\bar{b}$ $3P$ removed), floor 5 (theory floor 5 MeV), κ fixed at 0.3 or 0.5.

K	λ	variant	$\chi^2_{\rm data}$	$\chi^2_{\rm prior}$	$\sigma_V(0.3)$	$\sigma_V(1)$	$\sigma_V(3)$	$\sigma_V(6)$	$V(0.3)$	$V(3)$	$V(6)$
6	1	–	40.6	2.2	83	0	24	33	−2.204	−0.412	+0.149
8	0.1	–	35.6	2.5	280	0	46	51	−2.023	−0.477	+0.092
8	1	–	41.5	2.0	127	0	29	36	−2.197	−0.404	+0.157
8	1	floor 5	150.4	10.3	100	0	22	23	−2.113	−0.442	+0.123
8	1	$\kappa = 0.3$	46.6	2.1	143	0	29	36	−2.068	−0.402	+0.158
8	1	$\kappa = 0.5$	51.0	3.9	148	0	29	36	−1.968	−0.401	+0.160
8	1	no $3P$	31.7	2.4	149	0	35	76	−2.311	−0.407	+0.211
8	1	no D	20.8	1.6	127	0	31	38	−2.235	−0.408	+0.137
8	10	–	46.6	2.3	78	0	21	28	−2.116	−0.370	+0.193
8	100	–	51.3	7.4	58	0	17	21	−2.017	−0.338	+0.220
10	1	–	40.1	1.7	158	1	40	53	−2.133	−0.362	+0.211
12	1	–	40.4	1.3	124	1	37	47	−2.131	−0.363	+0.201

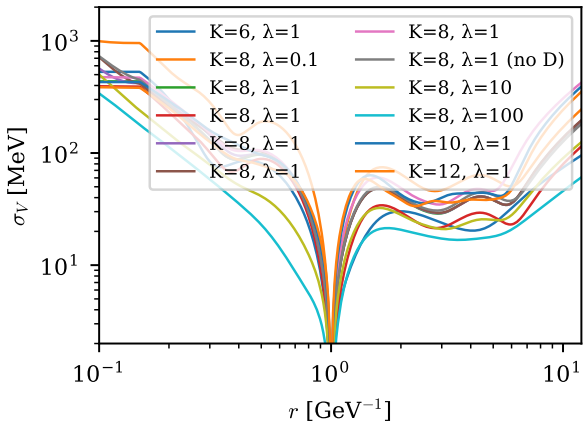


Figure 2: Width of the 1σ band for all fits in the stability scan. The dip at $r = 1\,\mathrm{GeV}^{-1}$ is the anchoring convention.

$b\bar{b}$ hyperfine splittings found in Ref. [8] with parametric potentials, now isolated to the one region of $V(r)$ that the spin-averaged spectrum does not constrain. The short-distance form of the potential is therefore an input, not an output, of this analysis; we carry $\kappa \in [0, 0.5]$ as a systematic.

Comparison with parametric forms. Because an additive constant is absorbed by the quark masses, parametric potentials must be compared at equal shape: in Fig. 1 and Table 1 each has been shifted to $V(1\,\mathrm{GeV}^{-1}) = -1.166\,\mathrm{GeV}$. So compared, the Martin and logarithmic potentials lie within 1–1.5σ of the extracted band at every r between 0.3 and $8\,\mathrm{GeV}^{-1}$; their small difference in curvature is below the resolution of the data. The Cornell potential is different: its linear rise is too steep at intermediate distances, $V(3\,\mathrm{GeV}^{-1}) = -0.498\,\mathrm{GeV}$ against the extracted -0.404 ± 0.029, a 3.2σ deviation that persists (-0.408) when the D waves are removed, and its Coulomb term is too strong at $0.2\,\mathrm{GeV}^{-1}$ (2.3σ). This is the quantitative version of the long-standing observation that a linear-plus-Coulomb form fits the heavy quarkonia less well than an effective power $r^{0.1}$ or a logarithm [3, 9]: the data require the slope of the confining part to fall with distance, from $dV/dr = 0.45\,\mathrm{GeV}^2$ at $1.5\,\mathrm{GeV}^{-1}$ to $0.25\,\mathrm{GeV}^2$ at $3\,\mathrm{GeV}^{-1}$ and $0.13\,\mathrm{GeV}^2$ at $6\,\mathrm{GeV}^{-1}$, whereas a linear term has a constant slope and must compromise between the two ends.

Stability. Table 2 summarises the scan. Increasing λ from 0.1 to 10 changes $V(3\,\mathrm{GeV}^{-1})$ by 0.1 GeV and χ^2 from 36 to 47; at $\lambda = 100$ the prior term ($\chi^2_{\rm prior} = 7$) is no longer negligible and the fit is over-regularised, so we exclude it. Within $0.1 \le \lambda \le 10$ the band widths at $r \ge 1\,\mathrm{GeV}^{-1}$ vary by less than a factor of two and the central values by less than one band width. Removing the D waves or the $3P$ levels moves $V(r)$ by at most 0.02 GeV anywhere; a 5 MeV floor sharpens the band and moves nothing. The extraction is robust in the range the data cover.

4 Consequences for B_c

The point of a potential with a covariance is that predictions inherit one. Table 3 gives the B_c levels obtained by solving the extracted potential for the $c\bar{b}$ system with the fitted m_c, m_b and Λ, together with two uncertainties: the propagated fit covariance, and the spread over the stability scan, which measures the residual dependence on K, λ, the floor, the data subset and κ. Both are small. Every level up to the $1D$ is predicted to ±4–5 MeV, with a variant spread of 2–4 MeV; the $3\,^3S_1$, whose wave function extends beyond $8\,\mathrm{GeV}^{-1}$ into the region where the potential is extrapolated, is the first level to inherit a large uncertainty ($\pm 17 \pm 25$ MeV), which is as it should be. The "model dependence" of 15–35 MeV quoted in Ref. [8] from the spread of three potentials was, for the lower levels, an overestimate by a factor of five: the spectrum determines the potential well enough that the choice of parametric form does not matter below the $3S$.

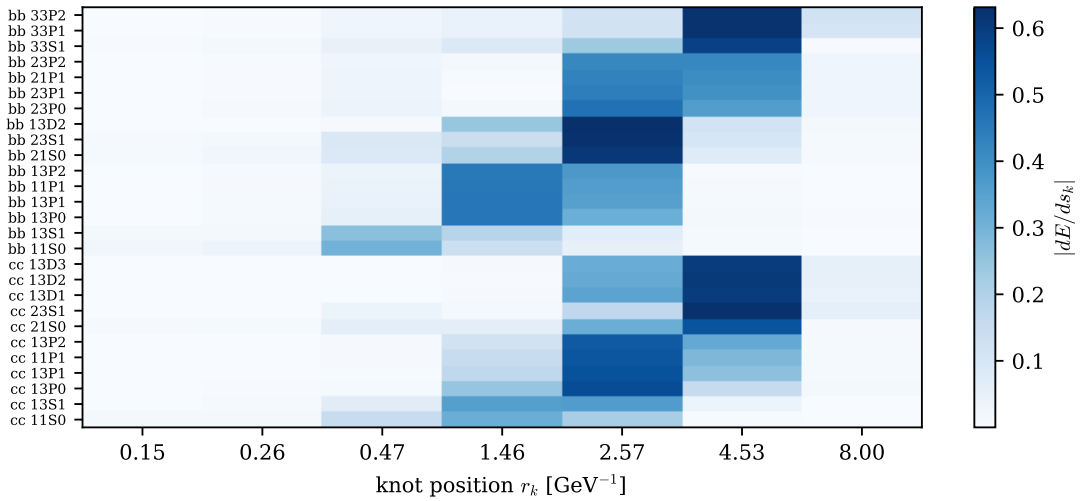


Figure 3: Sensitivity of each fitted level to the free knot values, $|\partial E_i/\partial s_k|$, for the reference fit. Rows are the 27 levels (charmonium below, bottomonium above), columns the knots.

Table 3: B_c masses (MeV) from the extracted potential. First error: propagated fit covariance (data plus prior); second: half-range over the eleven stability-scan fits of Table 2 with $\lambda \leq 10$. Experimental values from Refs. [10–12]. Our $1P$, $1D$ are the $2P$, $3D$ of principal-quantum-number labelling.

State	prediction	experiment
$1\,^1S_0$	$6264 \pm 4 \pm 3$	6274.5 ± 0.3
$1\,^3S_1$	$6336 \pm 5 \pm 4$	6339.0 ± 1.7
$2\,^1S_0$	$6874 \pm 4 \pm 2$	6871.2 ± 1.0
$2\,^3S_1$	$6912 \pm 4 \pm 3$	–
$1\,^3P_0$	$6711 \pm 4 \pm 4$	–
1^+	$6744 \pm 4 \pm 2$	–
$1^{+\prime}$	$6754 \pm 3 \pm 2$	–
$1\,^3P_2$	$6767 \pm 4 \pm 4$	–
$1\,^3D_1$	$7025 \pm 4 \pm 3$	–
$1\,^3D_3$	$7026 \pm 3 \pm 3$	–
$3\,^3S_1$	$7255 \pm 17 \pm 25$	–
Δ_{1S}	$71.2 \pm 3.7 \pm 6.7$	64.5 ± 1.7
Δ_{2S}	$38.4 \pm 2.5 \pm 4.0$	35.4 ± 2.4
$M(2\,^1S_0) - M(1\,^1S_0)$	$609 \pm 6 \pm 3$	596.7 ± 1.0

The predictions agree with experiment where it exists, with one exception that we think is significant. The $B_c^*(1S)$ at 6336 ± 6 MeV and the $B_c(2S)$ at 6874 ± 5 MeV match the ATLAS [11] and PDG values within errors, and the two hyperfine splittings, 71 ± 8 and 38 ± 5 MeV, agree with the ATLAS and ATLAS+CMS values at 1σ. The ground state, however, is predicted at 6264 ± 5 MeV, 10 MeV below the measured 6274.5 MeV, and consequently the $2S$–$1S$ spacing of B_c is 609 ± 7 MeV against the measured 596.7 ± 1.0: a 12 MeV, 1.8σ excess. In Ref. [8], with parametric potentials, the same spacing was 610–616 MeV; there it could be attributed to the potential. Here it cannot: the potential is whatever the $c\bar{c}$ and $b\bar{b}$ data require, and those data fix the spin-averaged $2S$–$1S$ spacings of charmonium (605 MeV) and bottomonium (572 MeV) to within the theory floor. For any flavour-independent central potential the $c\bar{b}$ spacing interpolates between them in reduced mass, and the extracted potential puts the spin-averaged $c\bar{b}$ spacing at 585 MeV; the measured value, 575 MeV using the ATLAS $2S$ hyperfine splitting, sits at the bottomonium end of the range. In the pseudoscalar channel the same 10–12 MeV appears as the 609 versus 597 MeV of Table 3.

The size and sign of the effect are what one expects from the leading correction to a flavour-independent static potential. In the $1/m$ expansion of the heavy-quark interaction [13] the first correction is $V^{(1)}(r)/m$ with a flavour-dependent coefficient $(1/m_c + 1/m_b)/2$ for $c\bar{b}$ versus $1/m_c$ for $c\bar{c}$ and $1/m_b$ for $b\bar{b}$; a term that is absorbed into $V^{(0)}$ for the equal-mass systems is not absorbed correctly for the unequal-mass one, and the mismatch is of order $(1/m_c - 1/m_b)^2\langle V^{(1)}\rangle$—tens of MeV. A 12 MeV shift between the $1S$ and $2S$ of B_c is therefore not a failure of the extraction but a measurement of the flavour dependence that the static potential omits. Its determination with the present precision has only become possible because the $B_c(2S)$ is now known to 1 MeV and the potential to a few MeV.

5 Conclusions

We have let the charmonium and bottomonium spectra determine the heavy-quark potential, and found:

1. $V(r)$ is known from the spectrum alone to 30–50 MeV for $0.7 \leq r \leq 7$ GeV^{-1} and to 0.1–0.3 GeV for $0.2 \leq r \leq 0.5$ GeV^{-1}; it is unconstrained below 0.2 GeV^{-1}, where only the hyperfine splittings act, and above 8 GeV^{-1}, where no bound state reaches.

2. The extracted shape fits the 27 levels with $\chi^2 = 40$, better than any parametric form; it is consis-

tent with a logarithmic or $r^{0.1}$ confinement and excludes the linear rise of the Cornell potential at $3\,\mathrm{GeV}^{-1}$ at 3σ.

3. Propagated to B_c, it gives predictions with ± 4–$5\,\mathrm{MeV}$ errors that do not depend on any choice of potential; the previously quoted model dependence of the low-lying B_c levels was overestimated by a factor of five.

4. The one discrepancy, a B_c $2S$–$1S$ spacing $12 \pm 7\,\mathrm{MeV}$ larger than measured while the $1S$ hyperfine, $2S$ hyperfine and $B_c(2S)$ mass agree, cannot be removed by changing $V(r)$ and is of the size expected for the flavour-dependent $1/m$ correction to the static potential.

The method is general: with the same code any future measurement—a B_c P-wave state, a $b\bar{b}$ D wave—can be added to the data set and will tighten the band in the region of r to which it is sensitive, which Fig. 3 tells in advance.

Acknowledgments

This work extends the author's MSc project at the University of Glasgow, supervised by Christine Davies, whose guidance is gratefully acknowledged. The calculations were performed on a single CPU; no high-performance computing resources were used. An AI assistant (Claude, Anthropic) was used to implement the spline fit and the analysis scripts and to help draft the manuscript; the author takes full responsibility for the physics, the code and the results.

Code and data availability

The code, all fits in the stability scan and the scripts that regenerate every table and figure are available at `https://github.com/apoppong24/bc-potential-inverse`; the experimental inputs are those of Ref. [8].

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